

Signal over substance: LLM research judges reward novelty rhetoric, and the bias resists steering

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Abstract

Caveats first. Our judges are three small open models (3.8–8B) and our ideas are *synthetic*; the effect’s magnitude is partly readout-dependent (it attenuates to borderline for one judge in a pairwise readout); and the near-perfect “detectability” of the bias is partly expected because the manipulation is surface-lexical. With that scope stated:

We ask whether an LLM used as a judge for research ideas scores *substantive novelty* (distance from prior work) or *surface novelty-signal* (rhetoric that asserts novelty). We build a correctness-controlled 2×2 design over 154 stems (616 items) crossing substance S (incremental vs. genuine departure, validated by a blind adjudicator) with signal G (plain vs. novelty-rhetoric, same technical content). Across all three judges the signal effect exceeds the substance effect (pooled $\beta_G = 1.03$ vs. $\beta_S = 0.67$), and in 0.82 of matched stems a low-substance idea in novelty rhetoric outscores a genuinely novel idea stated plainly. The effect is *not* explained by length. The signal response is a linear, domain-general direction in the judge’s residual stream (in-distribution AUROC ≈ 1.0 , out-of-domain ≥ 0.997)—yet projecting that direction out, at one layer or many, *does not recalibrate* the judge: β_G survives or β_S collapses with it. Detectability does not imply removability—convergent with prior work finding judge-bias steering unstable and multi-directional. Finally, one mechanism—scoring G not S (signal explains 0.63 of the score variance vs. 0.20 for substance)—reconciles two 2026 benchmarks that separately reported over-rating and score/substance decoupling. If an LLM judge is the verifier in an automated-research loop, it rewards sounding novel over being novel, and the bias cannot be cheaply patched out.

1 Introduction

A recurring thesis holds that the binding constraint on automated AI research is *verification*, not generation: models propose plausible research directions readily, but scoring which are actually good is the bottleneck, and LLM-as-judge is the proposed verifier. A judge is only useful if it rewards real quality. We test the specific failure mode most corrosive to research evaluation: rewarding the *rhetoric* of novelty rather than novelty itself. If a judge does that, then (i) it is a poor verifier, and (ii) placing it in a reinforcement loop trains generators to emit novelty-signalling language, not novel work—reward hacking of an absent oracle.

Contributions. (1) A correctness-controlled 2×2 that isolates surface novelty-signal from substantive novelty (§3); (2) the finding that all three judges score signal over substance, robust to length (§4); (3) a minimal reward-hack demonstrating the score is farmable with substance fixed (§5); (4) a mechanistic result: the signal response is a linear, OOD-generalizing direction that

is nonetheless *not* removable by activation steering (§6)—detectability $\neq$ steerability; and (5) a single-mechanism reconciliation of two 2026 benchmarks (§7).

2 Related work and how we differ

Novelty judging. RQ-Bench [1] finds LLM judges over-rate the novelty of model-generated research questions (a “novelty mirage”); RINoBench [2] finds their novelty verdicts diverge from expert gold with a regression-to-the-mean profile (never “not novel”, rarely top-grade). Both are observational; neither isolates novelty from correctness, nor localizes or intervenes on a mechanism. NovBench [3] finds LLMs have only limited understanding of novelty (it does not itself frame a surface/substance split). We supply the controlled crossed design and the mechanism, and reconcile the two benchmarks (§7). **Judge shortcuts.** Judges follow non-quality cues (source, verbosity, confidence, recency) while rationalizing as content [4, 5]; persuasion [6] and self-play [7] inflate scores without correctness. We port the cue-invariance paradigm to *novelty* with correctness fixed by construction, and demonstrate the novelty-specific hack. **Steering judge biases.** “Breaking the Mirror” [8] reduces unjustified self-preference via activation steering (up to 97%) but reports the steering is *unstable* and that self-preference spans multiple or nonlinear directions; [9] steers judges toward honesty; a creativity direction can be added to amplify creativity [10]. Our probe+direction machinery is the same, and our E4 result is *convergent* rather than contrary: the novelty-signal direction is linearly detectable yet not cleanly steerable away. We extend the “judge-bias steering is unstable” finding to a new bias and state the detectability/steerability dissociation explicitly.

3 The signal–substance design

Constructs. For each stem (a fixed prior-work context) we author an incremental idea (S =low) and a genuine departure (S =high), matched on plausibility; each is written plainly (G =low) and in novelty rhetoric (G =high) with *identical technical content*. This yields four cells LL, LH, HL, HH. **Validation.** An independent adjudicator, blind to framing, must agree which content departs more (S); all cells must be plausible and balanced with no content leak; and the signalled cells must out-score the plain cells on a judge-independent signal measure (a frozen lexicon and a MiniLM difference-of-centroids). This yields 154 validated stems (616 items) across ML and economics. **Judges & readout.** Phi-3.5-mini, Qwen2.5-7B, Llama-3.1-Nemotron-8B (4-bit). The novelty score Y is the expected digit over the next-token distribution restricted to 1–9 under a frozen rubric that asks for novelty only. We fit $Y \sim S + G + S:G$ with a per-stem random intercept; CIs are stem cluster bootstraps.

4 Judges score signal over substance

Table 1 and Fig. 1: for every judge $\beta_G > \beta_S$ with CIs excluding zero. Pooled $\beta_S = 0.67$, $\beta_G = 1.03$ (ratio 0.65). The *hackability index* $P(Y_{LH} > Y_{HL})$ —a low-substance signalled idea beating a high-substance plain one—is 0.82 pooled (0.89/0.85/0.71 per judge). Judges do track substance ($\beta_S > 0$), but signal dominates.

Not a length artifact (E1). Signalled cells are only $\sim 6\%$ longer; controlling for length within stem, β_G does not shrink (change $-3/-4/-6\%$; the length coefficient is near zero/negative).

Judge	β_S	β_G	ratio S/G	hackability
Phi-3.5-mini	0.40	0.75	0.54	0.89
Qwen2.5-7B	1.14	1.49	0.76	0.85
Nemotron-8B	0.47	0.84	0.55	0.71
Pooled	0.67	1.03	0.65	0.82

Table 1: E0. Signal effect β_G exceeds substance effect β_S for all judges.

LLM research judges score novelty SIGNAL over SUBSTANCE (N=154 stems, 616 items)

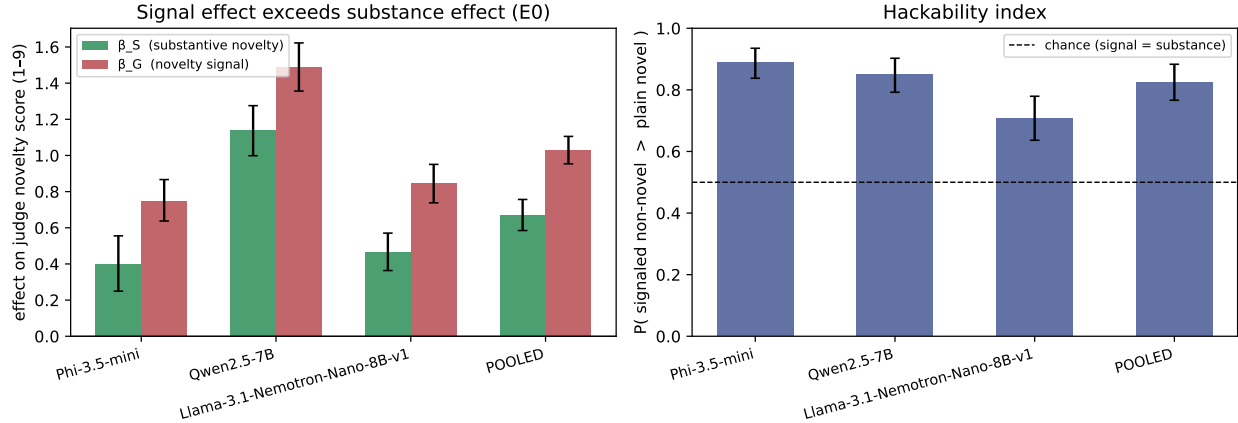


Figure 1: E0: signal beats substance (left) and the hackability index (right).

Survives comparison (E1). In a pairwise readout with position bias controlled, P (pick signalled non-novel) is 0.70 (Phi) and 0.55 (Nemotron)—both above chance—but 0.53 for Qwen (CI 0.48–0.58, borderline). So the effect is not merely a pointwise-readout artifact, though its magnitude is readout-dependent.

5 The score is reward-hackable

A minimal attack—greedily prepending pure-rhetoric sentences (adding no technical content) to a *plain* idea and keeping edits that raise Y —farms the score with substance fixed. On low-substance ideas the gamed gain is 0.99/2.10/1.92 points, and the fraction of low-substance ideas farmed to exceed the judge’s mean score on genuinely high-substance plain ideas rises from 47/7/20% to 100/83/100%.

6 Detectable but not steerable

A linear signal direction (E3). A logistic probe on the scoring-position residual separates G -high from G -low at in-distribution AUROC ≥ 1.000 for all judges, and *across domains* (train ML, test econ) at AUROC ≥ 0.997 . (Near-perfect separability is partly expected: the signal is surface-lexical.) **The fix fails (E4).** Projecting the difference-of-means signal direction out of the residual—at the best probe layer, or across an early-mid band—does not selectively remove the bias (Fig. 2): for Phi β_G is untouched (0.75→0.77) while β_S falls (0.40→0.19); for Nemotron both

collapse together (β_G 0.84 $\rightarrow$ 0.31, β_S 0.47 $\rightarrow$ 0.13); for Qwen the change is negligible. No judge shows the target ($\beta_G \downarrow$, β_S preserved). The novelty-signal response is entangled with novelty computation, not a cleanly separable direction—echoing prior work that found self-preference steering unstable and multi-directional [8].

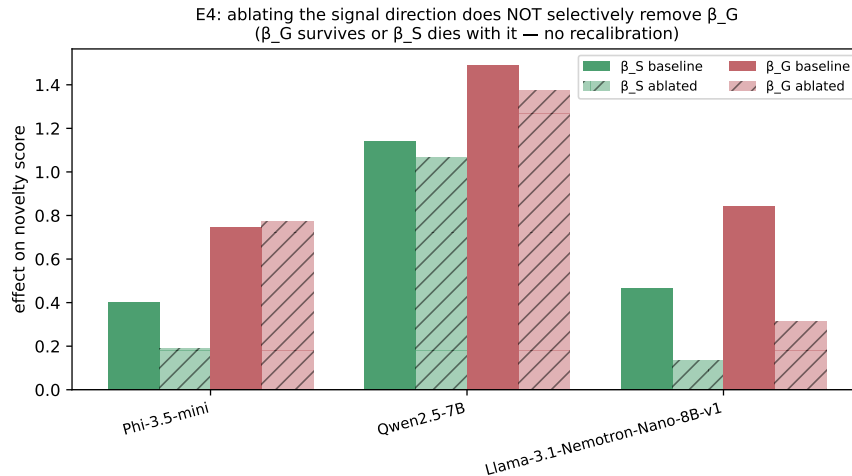


Figure 2: E4: ablating the signal direction does not selectively remove β_G .

7 One mechanism, both benchmarks

If the judge scores G not S , both prior findings follow. The score’s variance is dominated by signal (pooled share 0.63 for G vs. 0.20 for S ; per judge G 0.65/0.46/0.76). Non-novel ideas out-score the average novel idea 0.57/0.33/0.31 of the time (the over-rating of [1]), while genuine novelty framed plainly is under-credited (the decoupling of [2]). External validation on those corpora is future work (Fig. 3).

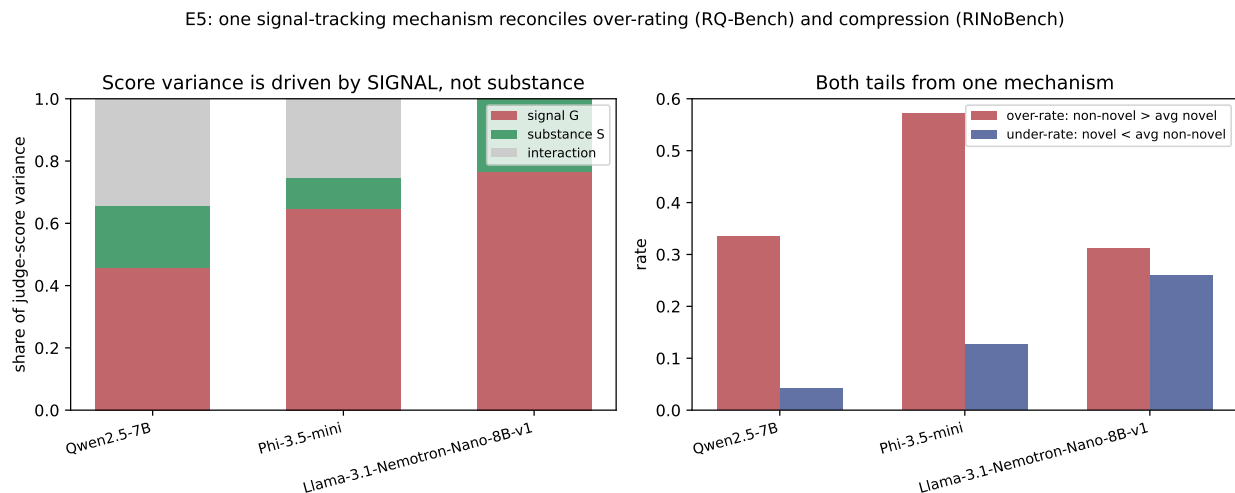


Figure 3: E5: signal dominates the score variance, producing both tails.

8 Discussion and limitations

The practical message is a warning: an LLM judge used as the verifier for automated research rewards sounding novel over being novel, the reward is farmable with substance fixed, and—like self-preference bias in prior work—it resists clean activation-steering removal. **Limitations.** Small open judges and synthetic ideas; the pairwise attenuation for Qwen shows readout-dependence; near-perfect probe AUROC is partly a surface-lexical artifact, so the load-bearing mechanistic claim is the *failure* of the causal fix, not the detection; the reconciliation is internal (controlled) evidence, not a re-analysis of the RQ-Bench/RINoBench corpora; and we test difference-of-means steering, not fine-tuning. Whether frontier judges, natural ideas, and stronger interventions change the picture is open.

References

- [1] *On the Limits of LLM-as-Judge for Scientific Novelty Assessment* (RQ-Bench), arXiv:2606.12071.
- [2] *Is this Idea Novel? An Automated Benchmark for Judgment of Research Ideas* (RINoBench), arXiv:2603.10303.
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- [10] *Steering Large Language Models to Evaluate and Amplify Creativity*, arXiv:2412.06060.